

ASSIGNMENT : C02-AT1

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Question 1:

1.A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

Genetic Algorithm to Maximize $f(x) = x^2$

Given :

Objective Function: $f(x) = x^2$

Initial Population: {2, 4, 6, 8}

Fitness Evaluation :

Individual (x)	Binary	Fitness $f(x) = x^2$
2	0010	4
4	0100	16
6	0110	36
8	1000	64

Selection:

Select the individuals with the highest fitness values.

Selected Parent	Individual	Fitness
Parent 1	8	64
Parent 2	6	36

Crossover:

Binary representation:

- Parent 1=1000
- Parent 2=0110

Exchange the least significant(last)bit.

Parent 1	Parent 2	Offspring 1	Offspring 2
1000	0110	1000(8)	0110(6)

Result: Since both parents have the last bit 0,the offspring remain unchanged.

Mutation:

Mutation randomly flips one bit.

Example:

Before Mutation	After Mutation	Decimal Value	Fitness
1000	1001	9	81

Role of Mutation :

- Introduces new genetic information into the population.
- Increases population diversity.
- Prevents premature convergence to local optima.
- Improves exploration of the search space.
- Helps the algorithm find better solutions.

Final Result :

Fitness values:

2 → 4

4 → 16

6 → 36

8 → 64

Selected parents: 8 and 6

Offspring after crossover: 8 and 6

Example after mutation: 9

Best fitness after mutation: 81

Conclusion: The Genetic Algorithm evaluates the fitness of each individual, selects the fittest parents, performs crossover to generate offspring, and applies mutation to improve diversity and increase the chances of finding the optimal solution.

Question 2:

2.A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is -0.4 . For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not.

INTRODUCTION :

A Perceptron is the simplest type of artificial neural network used for binary classification problems. It takes input values, multiplies them by their corresponding weights, adds a bias, and then applies an activation function to produce the output. The output is either 1 (Yes) or 0 (No).

Given Data :

Parameter	Value
Advertisement Exposure(x_1)	2
Income Level(x_2)	1
Weight(w_1)	0.5
Weight(w_2)	0.3
Bias(b)	-0.4

Step 1: Calculate the Net Input

The formula for net input is :

$$\text{Net Input} = (w_1x_1) + (w_2x_2) + b$$

Substitute the given values :

$$= (0.5 \times 2) + (0.3 \times 1) + (-4)$$

$$= 1.0 + 0.3 - 0.4$$

$$= 0.9$$

$$\text{Net Input} = 0.9$$

Step 2: Apply the Step Activation Function

The step activation function is :

- If $\text{Net Input} \geq 0$, $\text{Output} = 1$
- If $\text{Net Input} < 0$, $\text{Output} = 0$

Since:

$$0.9 \geq 0$$

$$\text{Output} = 1$$

Step 3 : Result

Net Input	Output	Decision
0.9	1	Customer will purchase the product

Explanation :

The perceptron first calculates the weighted sum of the inputs and adds the bias. The obtained net input is 0.9, which is greater than zero. Therefore, the step activation function produces an output of 1. This indicates that the customer is predicted to purchase the product.

Conclusion :

The perceptron successfully classifies the customer based on the given features.

Net Input = 0.9

Activation Output = 1

Prediction: The customer will purchase the product.